Introduction

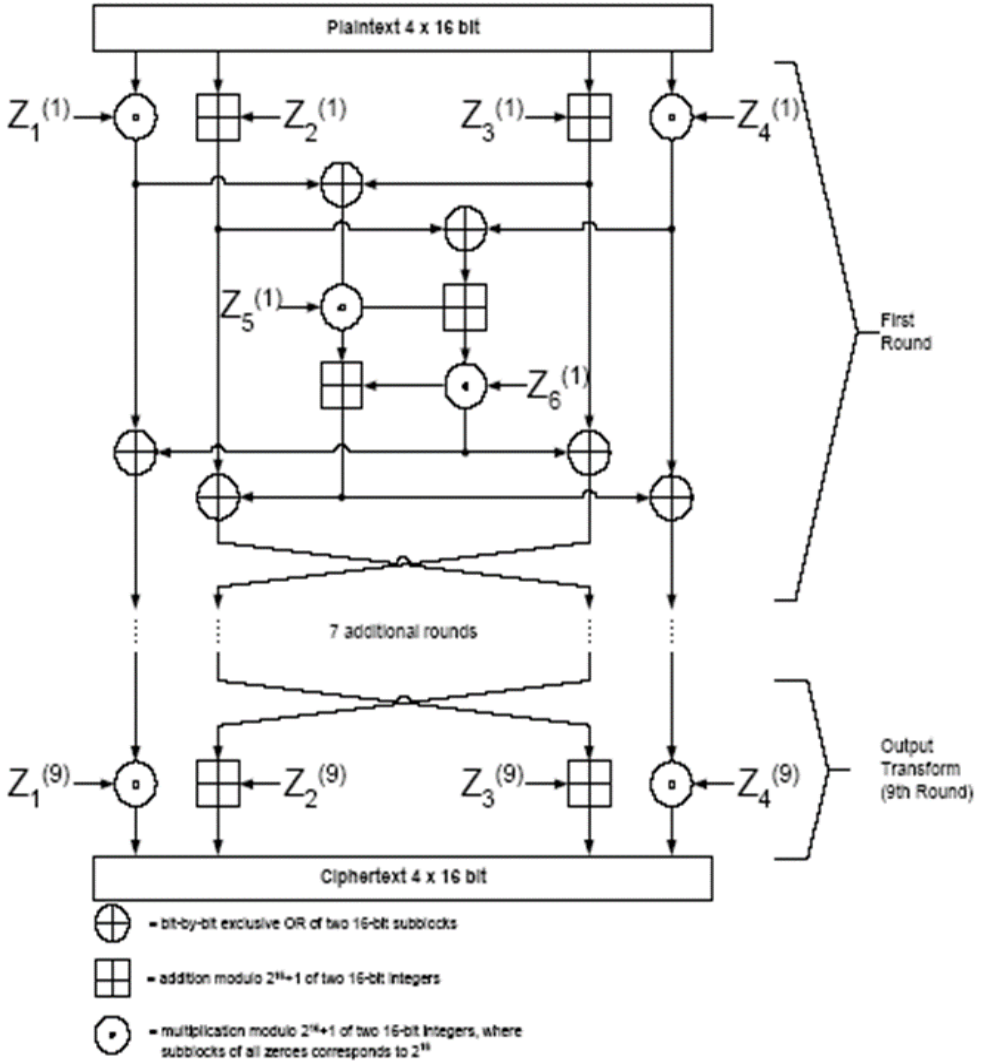
- DES algorithm has been a popular secret key encryption algorithm and is used in many commercial and financial applications.
- However, its key size is too small by current standards and its entire 56 bit key space can be searched in approximately 22 hours.
- IDEA is a block cipher designed by Xuejia Lai and James L. Massey in 1991.
- It is a minor revision of an earlier cipher, PES (Proposed Encryption Standard).
- IDEA was originally called IPES (Improved PES) and was developed to replace DES.
- It entirely avoids the use of any lookup tables or S-boxes.
- IDEA was used as the symmetric cipher in early versions of the Pretty Good Privacy cryptosystem.

Introduction … Contd.

- IDEA operates with 64-bit plaintext and cipher text blocks and is controlled by a 128-bit key.
- Completely avoid substitution boxes and table lookups used in the block ciphers.
- The algorithm structure has been chosen such that when different key sub-blocks are used, the encryption process is identical to the decryption process.

Key Generation

- The 64-bit plaintext block is partitioned into four 16-bit sub-blocks
- Six 16-bit keys are generated from the 128-bit key.
- Since a further four 16-bit key-sub-blocks are required for the subsequent output transformation, a total of 52 (= 8 x 6 + 4) different 16-bit sub-blocks have to be generated from the 128-bit key.



Key Generation … Contd.

- First, the 128-bit key is partitioned into eight 16-bit sub-blocks which are then directly used as the first eight key sub-blocks.
- The 128-bit key is then cyclically shifted to the left by 25 positions, after which the resulting 128-bit block is again partitioned into eight 16-bit sub-blocks to be directly used as the next eight key sub-blocks.
- The cyclic shift procedure described above is repeated until all of the required 52 16-bit key sub-blocks have been generated.

Encryption of the key sub-blocks

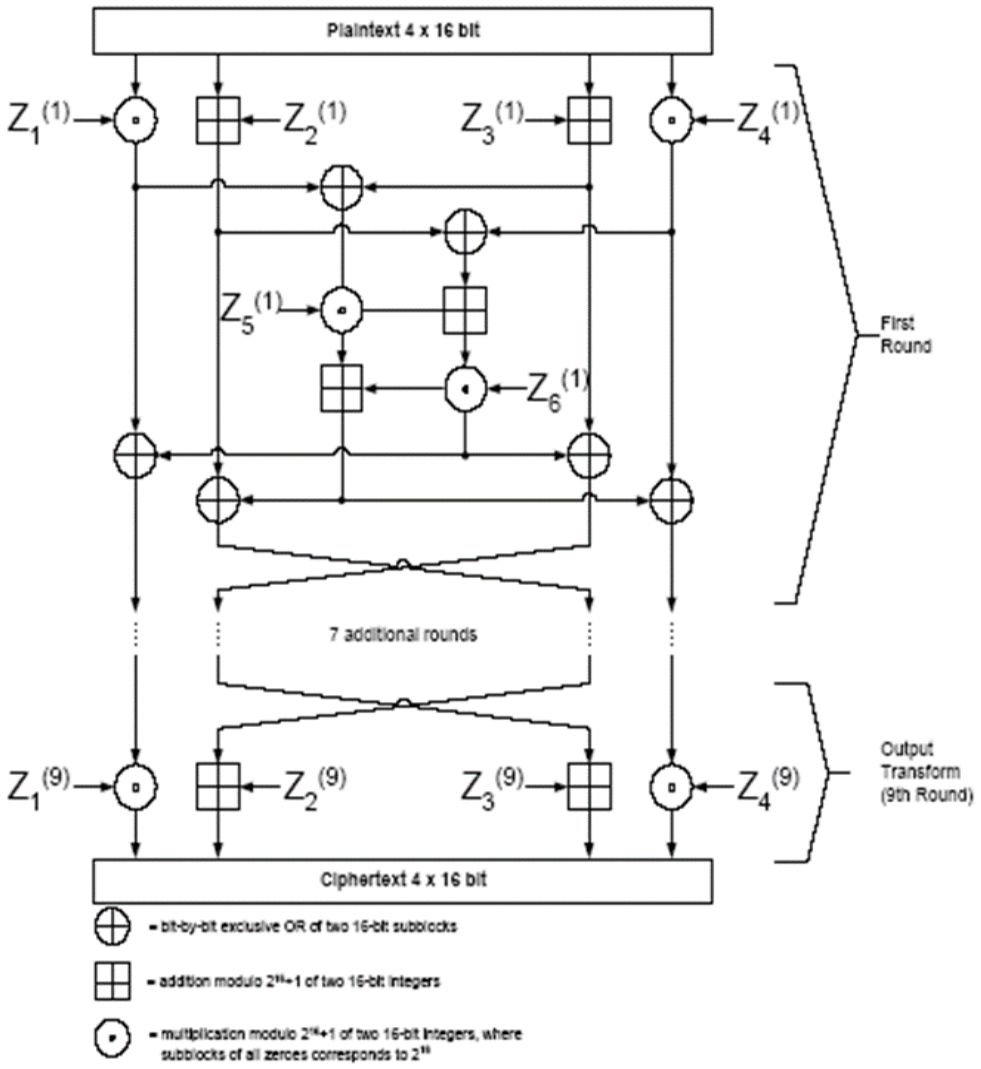
- The key sub-blocks used for the encryption and the decryption in the individual rounds.

Encryption of the key sub-blocks

Round 1	$Z_1^{(1)}\ Z_2^{(1)}\ Z_3^{(1)}\ Z_4^{(1)}\ Z_5^{(1)}\ Z_6^{(1)}$
Round 2	$Z_1^{(2)}\ Z_2^{(2)}\ Z_3^{(2)}\ Z_4^{(2)}\ Z_5^{(2)}\ Z_6^{(2)}$
Round 3	$Z_1^{(3)}\ Z_2^{(3)}\ Z_3^{(3)}\ Z_4^{(3)}\ Z_5^{(3)}\ Z_6^{(3)}$
Round 4	$Z_1^{(4)}\ Z_2^{(4)}\ Z_3^{(4)}\ Z_4^{(4)}\ Z_5^{(4)}\ Z_6^{(4)}$
Round 5	$Z_1^{(5)}\ Z_2^{(5)}\ Z_3^{(5)}\ Z_4^{(5)}\ Z_5^{(5)}\ Z_6^{(5)}$
Round 6	$Z_1^{(6)}\ Z_2^{(6)}\ Z_3^{(6)}\ Z_4^{(6)}\ Z_5^{(6)}\ Z_6^{(6)}$
Round 7	$Z_1^{(7)}\ Z_2^{(7)}\ Z_3^{(7)}\ Z_4^{(7)}\ Z_5^{(7)}\ Z_6^{(7)}$
Round 8	$Z_1^{(8)}\ Z_2^{(8)}\ Z_3^{(8)}\ Z_4^{(8)}\ Z_5^{(8)}\ Z_6^{(8)}$
Output Transform	$Z_1^{(9)}\ Z_2^{(9)}\ Z_3^{(9)}\ Z_4^{(9)}$

Encryption

- The first four 16-bit key sub-blocks are combined with two of the 16-bit plaintext blocks using addition modulo 2^{16}, and with the other two plaintext blocks using multiplication modulo $2^{16} + 1$
- At the end of the first encryption round four 16-bit values are produced which are used as input to the second encryption round
- The process is repeated in each of the subsequent 7 encryption rounds
- The four 16-bit values produced at the end of the 8th encryption round are combined with the last four of the 52 key sub-blocks using addition modulo 2^{16} and multiplication modulo $2^{16} + 1$ to form the resulting four 16-bit ciphertext blocks



Decryption

- The computational process used for decryption of the ciphertext is essentially the same as that used for encryption.
- The only difference is that each of the 52 16-bit key sub-blocks used for decryption is the inverse of the key sub-block used during encryption.
- In addition, the key sub-blocks must be used in the reverse order during decryption to reverse the encryption process.

Modes of Operation

- IDEA supports all modes of operation such as:
  - Electronic Code Book (ECB) mode
  - Cipher Block Chaining (CBC)
  - Cipher Feedback (CFB)
  - Output Feedback (OFB) modes

Applications of IDEA

- Today, there are hundreds of IDEA-based security solutions available in many market areas, ranging from Financial Services, and Broadcasting to Government
- The IDEA algorithm can easily be embedded in any encryption software. Data encryption can be used to protect data transmission and storage. Typical fields are:
  - Audio and video data for cable TV, pay TV, video conferencing, distance learning
  - Sensitive financial and commercial data
  - Email via public networks
  - Smart cards